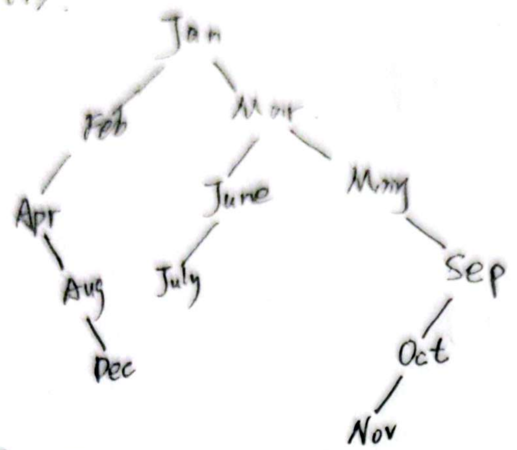
9.4

(1)



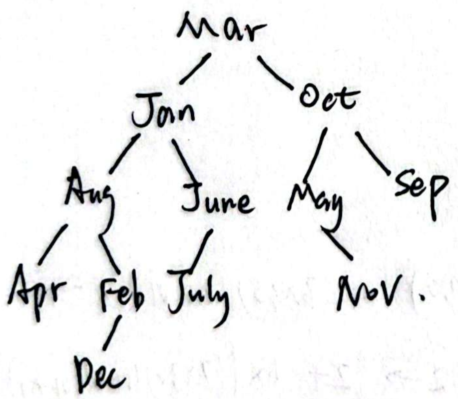
$$平均查找长度 = \frac{1\times1+2\times2+3\times3+4\times3+5\times2+6\times1}{12}$$

$$= 3.5$$

(2) 顺序表为 [Apr, Aug, Dec, Feb, Jan, July, June, Mar, May, Nov, Oct, Sep]

$$平均查找长度 = \frac{1\times1+2\times2+3\times4+4\times5}{12} \approx 3.08$$

(3).



$$平均查找长度 = \frac{1\times1+2\times2+3\times4+4\times4+5\times1}{12} \approx 3.17$$

扫描全能王 创建

9.14. [20, 30, 50, 52, 60, 68, 70]

① 插入20, 30.

[20, 30]

② 插入50, 发生上溢

[30]

[20] [50]

③ 插入52:

[30]

[20] [50, 52]

④ 插入60, 发生上溢

[30, 52]

[20] [50] [60]

⑤ 插入68. [30, 52]

[20] [50] [60, 68]

⑥ 插入70, 发生上溢.

[52]

[30] [68]

[20] [50] [60] [70]

删除50.

[52, 68]

[20, 30] [60] [70]

删除68.

[52]

[20, 30] [60, 70]

9.19. $H(k) = (3k) \pmod{11}$

$m = 11$

$H_i = (H(k) + i \times [(7k) \pmod{10} + 1]) \pmod{11}$

① 22.

$H(22) = (3 \times 22) (\bmod 11) = 0$

$SL = 1$

② 41

$H(41) = (3 \times 41) (\bmod 11) = 2$

$SL = 1$

③ 53.

$H(53) = (3 \times 53) (\bmod 11) = 5$

$SL = 1$

④ 46

$H(46) = (3 \times 46) (\bmod 11) = 6$

$SL = 1$

⑤ 30

$H(30) = (3 \times 30) (\bmod 11) = 2$

$2 \to 2 + 1 \times ((7 \times 30) (\bmod 10) + 1) = 3$

$SL = 2$

⑥ 13

$H(13) = (3 \times 13) (\bmod 11) = 6$

$step = (7 \times 13) \bmod 10 + 1 = 2$

$6 \to 6 + 1 \times 2 = 8$

$SL = 2$

⑦ 01

$H(1) = (3 \times 1) \bmod 11 = 3$

$step = 8$

$3 \to 0 \to 8 \to 5 \to 2 \to 10$

$SL = 6$

⑧ 67

$H(67) = 3$

$step = 10$

$3 \to 2 \to 1$

$SL = 3$

平均查找长度

$= \frac{1 \times 4 + 2 \times 2 + 3 \times 1 + 6 \times 1}{8}$

$= 2.125$

扫描全能王 创建

9.20. $\alpha = \frac{n}{m} \geq 0.75$

$\Rightarrow m \leq 16$

取 $m = 16, p = 13$

$H_i = (H(key) + i) \pmod{16}$

key：取首字母在字母表中的序号.

① ZHAO.

$H = 26 \pmod{13} = 0$

$SL = 1$

② QIAN

$H = 17 \pmod{13} = 4$

$SL = 1$

③ SUN.

$H = 19 \pmod{13} = 6$

$SL = 1$

④ LI

$H = 12 \pmod{13} = 12$

$SL = 1$

⑤ ZHOU

$H = 26 \pmod{13} = 0$

$0 \rightarrow (0+1) \bmod 16 = 1$

$SL = 2$

⑥ WU

$H = 23 \pmod{13} = 10$

$SL = 1$

⑦ ZHANG

$H = 26 \pmod{13} = 0$

$0 \rightarrow 1 \rightarrow 2$

$SL = 3$

⑧ WANG

$H = 23 \pmod{13} = 10$

$10 \rightarrow 11$

$SL = 2$

⑨ CHANG

$H = 3 \pmod{13} = 3$

$SL = 1$

⑩ CHAO

$H = 3$

$3 \rightarrow 4 \rightarrow 5$

$SL = 3$

⑪ YANG

$H = 25 \pmod{13} = 12$

$12 \rightarrow 13$.

$SL = 2$

⑫ JIN

$H = 10$

$10 \rightarrow 11 \rightarrow 12 \rightarrow 13 \rightarrow 14$

$SL = 5$

$$ASL = \frac{1 \times 6 + 2 \times 3 + 3 \times 2 + 5 \times 1}{12} \approx 1.92$$

9.21

哈希地址为[5,3,6,0,6,5,5,0,9,7,7,2]

1) 线性探测再散列地址法

~~回答如~~

地址	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
元素	Apr	Aug	Dec	Feb		Jan	Mar	May	June	July	Sep	Oct	Nov				
SL	1	2	1	1		1	1	2	4	5	2	5	6				

查找成功：$\frac{1+2+1+1+1+1+1+2+4+5+2+5+6}{12} \approx 2.58$

查找不成功：$\frac{5+4+3+2+1+9+8+7+6+5+4+3+2+1}{14} \approx 4.29$

(2) 链地址法

地址 0：Apr → Aug

1：^

2：Dec

3：Feb

4：^

5：Jan → June → July

6：Mar → May

7：Oct → Nov

8：^

9：Sep

10~16：^

查找成功：$\frac{(1+2)+1+1+(1+2+3)+(1+2)+(1+2)+1}{12} = 1.5$

查找不成功：$\frac{2+1+1+3+2+2+1}{14} \approx 0.86$

扫描全能王 创建